

CAIR Test Flow

ClaudMD / Aeliusmd → CAIR2 HL7 VXU | Sithum Dev DB | Org SF-013259

1. What We Done

- Built a Python CAIR integration that reads vaccination data from ClaudMD DB, builds HL7 VXU messages, and sends them to CAIR2 over SOAP (submitSingleMessage).
- Synced Sithum dev DB (ClaudMD_Development_Sithum) with QA_2 so all CAIR tables/columns match.
- Configured .env for Sithum dev using CLINIC_DB_* (direct clinic DB connection).
- Created scripts/seed_sithum_cair_test_data.py to insert test vaccines + queue rows for testing.
- Seeded 7 test submissions in Sithum DB; test sends attempted to CAIR onboarding endpoint.
- Fixed SOAP client to SOAP 1.2 per CAIR WSDL (was HTTP 500 with old SOAP 1.1 envelope).
- Org code from onboarding email is set: SF-013259. Stage/onboarding SOAP URL is set.
- Waiting on CAIR SOAP username/password (CAIR_SOAP_USERNAME / CAIR_SOAP_PASSWORD in .env).

2. How Background Job Runs

- Run **python cair_service.py** — one long-running process for all clinics.
- Every 5 minutes it wakes up, checks for pending work, processes a batch, then sleeps again.
- In dev (Sithum): connects directly to one clinic DB from CLINIC_DB_*.
- In QA/prod: reads ClinicSetup from master DB, then connects to each active clinic DB.
- For each pending row: read DB → build HL7 → send SOAP → update status in the same table.
- Up to 3 clinics can be processed in parallel inside that one process (WORKER_POOL_SIZE=3).

```
cair_service.py (every 5 min)
|
+-- Dev: CLINIC_DB_* -> ClaudMD_Development_Sithum
+-- QA:  Master DB -> ClinicSetup -> each clinic DB
|
Find SubmitStatus = 0 or 3 AND EHRHeaders.IsPublish = 1
|
Build HL7 VXU -> SOAP submitSingleMessage -> CAIR2
|
UPDATE EHRVaccineThirPartySubmissions + EHRVaccines
```

2.1 How Workers Work (3 Layers)

Layer	File	Role
1 — Scheduler	cair_service.py	Starts once, runs forever. Every SCHEDULER_INTERVAL_SECONDS (300s = 5 min) calls coordinator.run_once(), then sleeps.
2 — Coordinator	worker/coordinator.py (CairCoordinator)	Finds clinics with pending work. Dispatches each clinic to a thread in ThreadPoolExecutor (WORKER_POOL_SIZE=3).
3 — Processor	worker/processor.py (SubmissionProcessor)	Per clinic: loads up to WORKER_BATCH_SIZE (100) pending rows, processes each submission one by one.

One background job, not one per clinic. A single cair_service.py process handles all clinics. Inside it, up to 3 clinic batches run in parallel threads.

2.2 Each Run Cycle (coordinator.run_once)

```
cair_service.py
|
```

```

CairCoordinator.run_once()
|
[Dev] CLINIC_DB_* -> check has_eligible_work() -> 1 clinic
[QA] Master DB -> ClinicSetup (Active=1) -> check each clinic
|
ThreadPoolExecutor (max 3 workers)
/      |      \
Clinic A  Clinic B  Clinic C  (only clinics with pending rows)
batch    batch    batch

```

2.3 Per Clinic Batch (process_clinic_batch)

Step	What happens	DB / code
1	Connect to clinic DB	CairSubmissionRepository
2	SELECT up to 100 rows: SubmitStatus=0 or 3, IsPublish=1	get_eligible_submissions()
3	Set SubmitStatus = 4 (PROCESSING) on selected rows	claim_submissions()
4	For each row — process_submission()	SubmissionProcessor

2.4 Per Submission (process_submission)

Step	Action	Result
1	Load patient + vaccine from DB (JOIN Patients, EHRVaccines, etc.)	VxuPayload
2	Build HL7 VXU message	vxu_builder.py
3	POST SOAP 1.2 submitSingleMessage (username, password, facilityID, hl7Message)	soap_client.py
4	Parse ACK (MSA segment)	ack_parser.py — AA / AE / AR
5a	ACK = AA (accept)	SubmitStatus=1, save HL7+ACK, IsSubmitted=1
5b	HTTP error, ACK = AE, or timeout (temp error)	SubmitStatus=3, AttemptCount+1, retry later
5c	Missing SOAP credentials or ACK = AR / max retries (5)	SubmitStatus=2, save ErrorMessage

2.5 Retry Logic

Attempt	Wait before retry	After 5 failures
1	5 minutes	
2	15 minutes	
3	30 minutes	
4	60 minutes	
5	120 minutes	Mark FAILED (SubmitStatus = 2)

- Rows with SubmitStatus=3 (RETRY) are picked up again after the wait period on the next run cycle.

2.6 Worker Config (.env)

Setting	Default	Purpose
SCHEDULER_INTERVAL_SECONDS	300	Seconds between each run cycle (5 min)
WORKER_POOL_SIZE	3	Max clinics processed in parallel
WORKER_BATCH_SIZE	100	Max submissions per clinic per cycle

Setting	Default	Purpose
MAX_RETRY_ATTEMPTS	5	Retries before marking FAILED

3. CAIR Endpoints & DB (Sithum Dev Now)

CAIR SOAP Endpoints (Onboarding vs Production)

Onboarding / stage (used now): URL in .env (CAIR_SOAP_URL) on **cdph-interop-stage.cdph.ca.gov** — for SF-013259 testing only; HL7 is sent via **SOAP 1.2 submitSingleMessage**. **Production (later):** URL on **cdph-interop-prod.cdph.ca.gov** — switch after CAIR approves go-live.

Environment	Submission endpoint (CAIR_SOAP_URL)
Onboarding / stage (now)	https://cdph-interop-stage.cdph.ca.gov/services/client_Service.client_ServiceHttpSoap12Endpoint
Production (after approval)	https://cdph-interop-prod.cdph.ca.gov/services/client_Service.client_ServiceHttpSoap12Endpoint

SOAP 1.2 envelope (per WSDL)

Item	Value
WSDL (reference)	https://cdph-interop-stage.cdph.ca.gov/CASTG-WS/IISService?WSDL
SOAP version	1.2 (Soap12Endpoint — SOAP 1.1 returns HTTP 500)
Namespace	urn:cdc:iisb:2011
SOAP action	urn:cdc:iisb:2011:submitSingleMessage
Body fields	username, password, facilityID (SF-013259), hl7Message
.env credentials	CAIR_SOAP_USERNAME, CAIR_SOAP_PASSWORD (from CAIR onboarding)

What DB We Use

Item	Value
Server	10.103.0.211
Database	ClaudMD_Development_Sithum
Tables read	EHRVaccineThirdPartySubmissions, EHRVaccines, EHRHeaders, CheckInsHeader, Patients, ServiceCodes
Tables updated	EHRVaccineThirdPartySubmissions, EHRVaccines only
Worker inserts?	No — production worker never INSERTs (read + UPDATE only)

4. What Fields We Send (HL7 from DB)

Source table	Columns used in HL7
Patients	AccountNumber, FirstName, LastName, DOB, GenderId, HomePhone, CellPhone, Email, Address, City, State, Zip
CheckInsHeader	Id (check-in), CheckInDate, CheckInTime
EHRVaccines	LotNumber, VaccineDate, VaccineTime, Dosage, Manufacturer, Route, BodySite, VaccineExpirationDate
ServiceCodes	NDCNumber, Description
EHRHeaders	IsPublish = 1 (filter — only published visits)
.env config	SF-013259 in HL7 as org code (MSH-4, PID-3, RXA site, etc.)

5. What Fields We Update After Send

EHRVaccineThirdPartySubmissions

Column	Purpose
SubmitStatus	0 pending → 4 processing → 1 success / 2 failed / 3 retry
RequestPayload	HL7 message sent
ResponsePayload	CAIR ACK response
SubmittedDateTime	When CAIR accepted
LastAttemptDateTime	Last send attempt time
AttemptCount	Retry counter
ErrorMessage	Error text on fail/retry
ExternalReferenceId	MSH-10 message control ID

EHRVaccines

- **IsSubmitted = 1** on success only.

6. IsPublish = 1 — What It Means

- **EHRHeaders.IsPublish** = whether the visit/chart is published in ClaudMD (provider finalized the EHR).
- **IsPublish = 1** → visit is ready; CAIR worker may pick up vaccines for that check-in.
- **IsPublish = 0** → visit not published; worker skips it even if a queue row exists.
- **It does NOT mean “sent to CAIR.”** It only means the visit is ready to send.

7. DB Columns — Was CAIR Sent or Not?

Main column: EHRVaccineThirdPartySubmissions.SubmitStatus

Value	Meaning
0	Not sent yet (pending)
1	Sent successfully to CAIR

Value	Meaning
2	Failed permanently
3	Failed temporarily — will retry
4	Currently sending

Also on EHRVaccineThirdPartySubmissions:

- RequestPayload — HL7 sent (empty = not sent yet)
- ResponsePayload — CAIR ACK response
- SubmittedDateTime — when CAIR accepted it
- ErrorMessage — error if failed

Secondary flag: EHRVaccines.IsSubmitted

- 0 or NULL → vaccine not reported to CAIR yet.
- 1 → vaccine successfully sent to CAIR (worker sets on success).

```

EHRHeaders.IsPublish = 1          -> Visit is ready (gate before send)
+
EHRVaccineThirdPartySubmissions.SubmitStatus = 0 -> Queued for CAIR
|
Worker sends HL7
|
SubmitStatus = 1 + EHRVaccines.IsSubmitted = 1 -> Sent to CAIR

```

Quick rule: check **SubmitStatus** on EHRVaccineThirdPartySubmissions — that is the main “CAIR sent or not” column.

8. Status Right Now

Check	Current state
Test data in Sithum DB	Ready (7 submissions seeded)
SOAP envelope	Fixed — SOAP 1.2 per WSDL (HTTP 500 resolved)
CAIR SOAP credentials	Not set — need username/password from CAIR onboarding
Test sends to CAIR	Attempted; submissions 1 & 7 failed (no credentials yet)
Best test record	Submission 7 — KEVIN TEST, checkin 1622, NDC 58160082152
ACK = AA (success)	Not yet — blocked until credentials are added

9. Next Steps

- Email CAIRDataExchange@cdph.ca.gov — request onboarding SOAP username/password for org SF-013259.
- Add CAIR_SOAP_USERNAME and CAIR_SOAP_PASSWORD to .env.
- Reset failed rows: python scripts/seed_sithum_cair_test_data.py reset
- Retry send: python run_dry_once.py (or python cair_service.py)
- After ACK = AA, ask CAIR to review test submissions before production go-live.

10. HL7 Built from This Record (Sithum DB — submission Id 7)

```
MSH|^~\&|ClaudMD|SF-013259||CAIR2|...|VXU^V04^VXU_V04|...|P|2.5.1|||AL|AL|||Z22^CDCPHINVS|SF-013259
PID|1||100031^^^SF-013259^MR||TEST^KEVIN...|20030314|F|...|ADDRESS 1*^^MAYFAIR^NY^12302...
|^PRN^PH^^^123^1323123~^NET^Internet^sakvith@aeliusmd.com~^PRN^CP^^^213^1231231|...
PD1|||||Y|||||
ORC|RE||1622^SF-013259|1622^SF-013259|...
RXA|0|1|20260816||58160082152^^NDC|0.5|mL^^UCUM|...|^SF-013259|...|LOT-CAIR-TEST-003|...
OBX|1|CE|64994-7^...|1|V01^Not VFC eligible^HL70064|...|20260816
```

Each field — where it comes from

HL7 part	Field	Where it comes from
MSH	ClaudMD	.env / code — sending application name
MSH	SF-013259	.env SENDING_FACILITY_ID — your CAIR org code
MSH	CAIR2	.env RECEIVING_FACILITY — California registry
MSH	VXU^V04^VXU_V04	Fixed — vaccination update message type
MSH	P	.env PROCESSING_ID — P = production/training mode
MSH	AL AL	Fixed — always accept ack (per CAIR email)
MSH	Z22^CDCPHINVS	Fixed — immunization message profile
MSH	SF-013259 (MSH-22)	.env RESPONSIBLE_ORG_ID — site that gave vaccine
PID	100031^^^SF-013259^MR	Patients.AccountNumber + org code SF-013259
PID	TEST^KEVIN	Patients.LastName + Patients.FirstName
PID	20030314	Patients.DateOfBirth
PID	F	Patients.GenderId (2 = Female)
PID	ADDRESS 1*^^MAYFAIR^NY^12302	Patients.Address1, City, State, ZipCode
PID	^PRN^PH^^^123^1323123	Patients.HomePhone — home phone in PID-13
PID	^NET^Internet^sakvith@aeliusmd.com	Patients.Email — email in PID-13
PID	^PRN^CP^^^213^1231231	Patients.CellPhone — cell phone in PID-13
PD1	Y	Default in code — protection indicator
ORC	RE	Fixed — replace/update order
ORC	1622^SF-013259	EHRVaccines.CheckInId (= CheckInsHeader.Id) + org code
RXA	20260816	EHRVaccines.VaccineDate — vaccination date
RXA	58160082152^^NDC	ServiceCodes.NDCNumber — vaccine product code
RXA	0.5 mL^^UCUM	EHRVaccines.Dosage — dose amount and unit
RXA	^^SF-013259	.env RESPONSIBLE_ORG_ID — where shot was given
RXA	LOT-CAIR-TEST-003	EHRVaccines.LotNumber — vaccine lot number
OBX	64994-7^Vaccine funding...	Fixed — VFC eligibility observation code
OBX	V01^Not VFC eligible	Default in code — not VFC eligible
OBX	20260816	EHRVaccines.VaccineDate — observation date